

EU Drug Market: MDMA — Retail markets



How is MDMA sold in Europe?

MDMA is most commonly found on the European market in the form of ecstasy tablets, relatively pure crystals or powders (finely ground crystals of MDMA that are typically mixed with other substances). These products typically contain MDMA hydrochloride (see Box [Forms of MDMA present on the European market](#)).

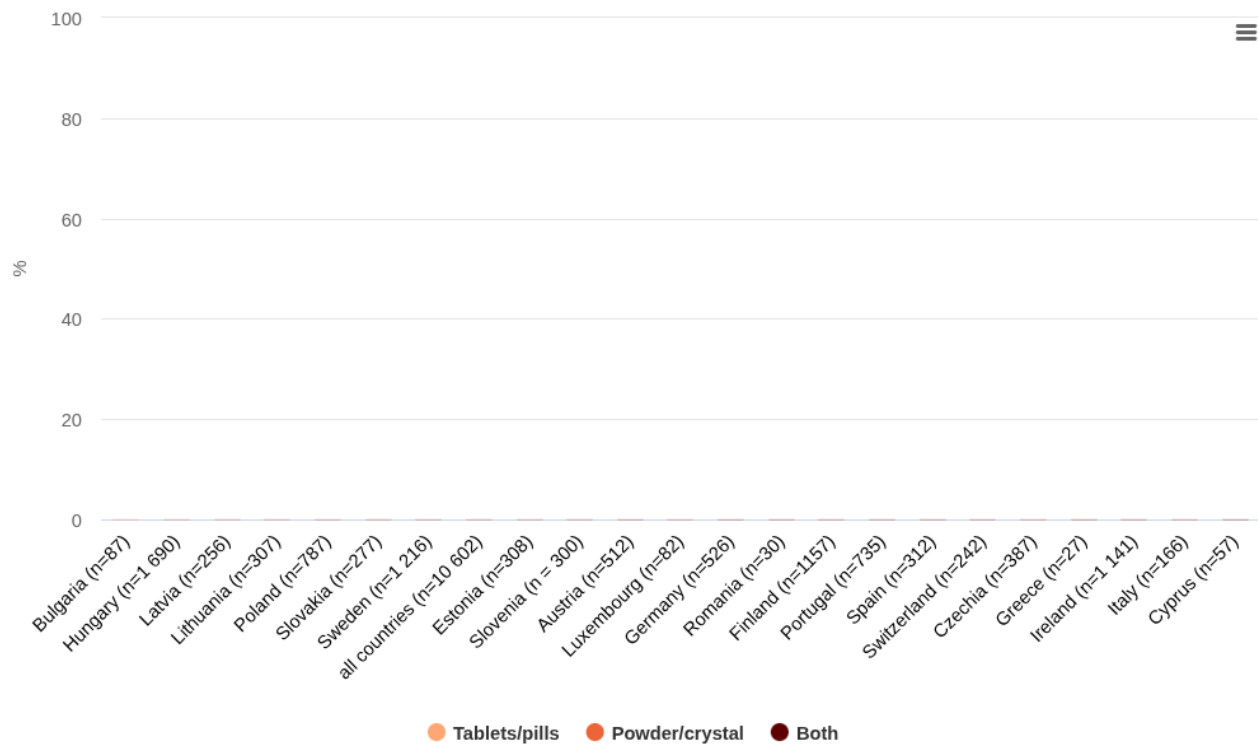
Since they emerged on the European drug market in the 1990s, ecstasy tablets were the most common form of MDMA consumed in Europe, but following a period where consumer trust in these products was low around the mid to late 2000s, powders/crystals gained a share of the market ([Gremeaux, 2022](#); [Mounteney et al., 2018](#)). According to 2021 data, tablets and powders/crystals share the EU MDMA market equally, with some variability among countries in terms of preference for either ecstasy tablets or MDMA powders/crystals (see [Figure 6.1. Distribution of usual MDMA forms across countries](#)). While not representative, data from the 2021 European Web Survey on Drugs (EWSD) (¹) found that tablets were used by 42 % of the 17 098 respondents who had used MDMA in the last 12 months, compared with 24 % for powder/crystal. Around 33 % of respondents reported using both forms of the drug.

This resource is part of [EU Drug Market: MDMA — In-depth analysis](#) by the EUDA and Europol.



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Figure 6.1. Distribution of usual MDMA forms across countries



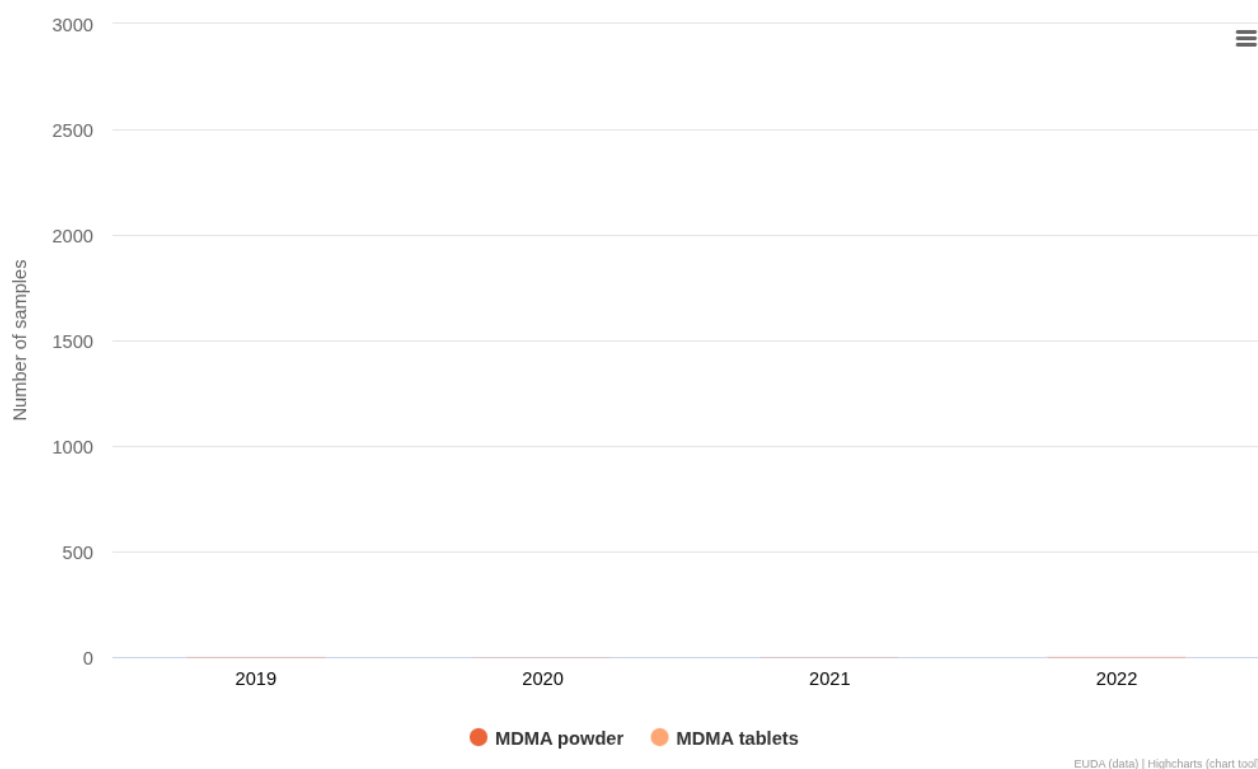
EUDA (data) | Highcharts (chart tool)

Source: European Web Survey on Drugs (EWSD), 2021. Survey question: 'What forms of ecstasy/MDMA do you usually use?'. See [Methodology](#).

Samples submitted to drug-checking services can also provide an indication of the products available to consumers. However, these data are not generalisable either, and as such should be interpreted with caution. Of the 6 811 samples submitted as MDMA to drug-checking services in eight EU countries (²) between 2019 and 2022, 57 % were tablets and 42 % were crystals (including 'rocks') or powders, without significant year-on-year variation (see [Figure 6.2. MDMA samples analysed by drug-checking services, 2019-2022](#)). Additional data were shared with the EUDA by the Drug information and Monitoring System (DIMS) in the Netherlands. Of nearly 53 500 MDMA samples analysed by DIMS over the same period, the average proportion of tablets was 58 % (ranging from 48 % in 2020 to 65 % in 2022).

Figure 6.2. MDMA samples analysed by drug-checking services, 2019-2022

MDMA samples analysed by drug-checking services



Source: TEDI (Trans-European Drug Information project).

MDMA samples analysed by drug-checking services by sample form, 2019-2022

● Crystal/powder/rock ● Capsule ● Tablet ● unspecified

EUDA (data) | Highcharts (chart tool)

Source: TEDI (Trans-European Drug Information project).

Other types of MDMA products occasionally appear on the drug market. For example, alcoholic drinks containing MDMA are submitted for drug checking from time to time. More rarely, almost pure MDMA base in liquid form has been sold to consumers, which represents a serious health risk (Drugs Information Monitoring System [DIMS], 2022). In recent years, the appearance of alternative (non-tablet) edible consumer products containing MDMA has also been reported, including candies, jellies, gummies and lollipops (see [Box 6.1. Lollipops containing MDMA seized in Belgium](#)). These products may make the use of MDMA more socially acceptable and appeal to non-habitual consumers and young people.

Box 6.1. Lollipops containing MDMA seized in Belgium

Dozens of lollipops containing MDMA were seized in 2021 in Belgium. The products had a net weight of approximately 20 grams each and contained between 130 and 176 milligrams of MDMA, evenly distributed inside the lollipops. Although unusual, this was not the first time that these types of products had been found in Europe. Sugary products containing MDMA may disguise the taste of the substance and mask the Marquis reagent test that is commonly used to identify the presence of MDMA in a sample.



Source: Belgian Federal Police.

MDMA is mostly bought face-to-face from known sellers

Studying the drug purchasing behaviours of users is useful for understanding market dynamics, but challenging given the clandestine nature of these transactions. Web-based surveys, such as the EWSD, can provide a helpful insight, although the results cannot be generalised to the population as a whole, or indeed to all people who use drugs. Data from drug-checking services can also inform on the type of relationships that exist between the consumer and the provider.

Together, these two data sources suggests that the large majority of MDMA users buy the drug themselves, and that a smaller proportion obtains it for free, despite marked differences among EU countries. Consumers appear to mostly buy MDMA from what they perceive to be trusted sources (usual suppliers, close friends and friends), but street suppliers and acquaintances are also relevant sources, as is the internet (including surface web, darknet and social media), albeit to a smaller extent.

In 2021, among the EWSD respondents who had used MDMA in the last 12 months ($n = 17\,098$), close to three quarters (72 %) reported mostly buying the drug, whereas one quarter (25 %) declared that they had obtained it for free. Differences between countries were apparent, however. For example, 79 % of the Polish respondents reported mostly buying the drug, compared

to 51 % of the respondents from Luxembourg.

Among EWSD respondents who provided further information ($n = 6\,946$), a clear majority (78 %) had bought MDMA directly from a dealer, in person, via phone, mail or text message. Smaller numbers of respondents also reported using the darknet (13 %) and social media (7 %). The use of these methods differed between countries, indicating that some purchasing practices are more common in some countries than in others. For example, the proportion of those buying MDMA directly from a dealer ranged from 61 % in Finland and Sweden to 95 % in Spain, and figures for the use of the darknet ranged from 0 % in Bulgaria and Cyprus to about 44 % in Finland. It should be noted, however, that the sample sizes were small in some countries.

More than three quarters of EWSD respondents (77 %) reported meeting their dealer outside as the usual method for receiving drugs, followed by home delivery (17 %). An equal percentage of respondents reported delivery through regular mail or post, and picking the drugs up from a location without any personal contact (7 %).

Data collected from individuals submitting 6 811 MDMA samples to drug-checking services between 2019 and 2022 appear to support these findings from the EWSD. Based on reporting from drug-checking clients, the majority of samples were reported as being obtained from a supplier (either a trusted supplier or a random street supplier) and a smaller proportion (close to 5 %) as bought on the internet (surface web or darknet).

A smaller share of retail consumers buys MDMA online

MDMA is one of the most frequently traded drugs online across a range of platforms, including darknet markets (see Section [Trafficking and distribution — online trade and distribution on darknet markets](#)) as well as surface web and social media platforms. Despite this, on average, only a minority of EWSD respondents reported obtaining the drug online, although this varied widely across countries. This appears to be supported by existing studies that indicate that MDMA is bought online mostly by retail distributors for the purpose of re-sale rather than purchased directly by retail clients ([Aldridge and Décary-Hétu, 2016](#); [Gremeaux, 2022](#)).

Recently, however, social media platforms and instant messaging applications have become particularly dynamic online channels through which MDMA is sold to consumers. The growing exploitation of these platforms is likely tied to the objective of reaching a wider audience. This is a development that requires closer monitoring.

Retail quantities sold online are often dispatched in post and parcels from EU Member States to consumer markets in Europe and globally, and cryptocurrencies are often used to settle these online transactions. Additionally, particularly with instant messaging applications, transactions may be organised online, but completed using in-person pick-up or 'dead-drop' delivery.

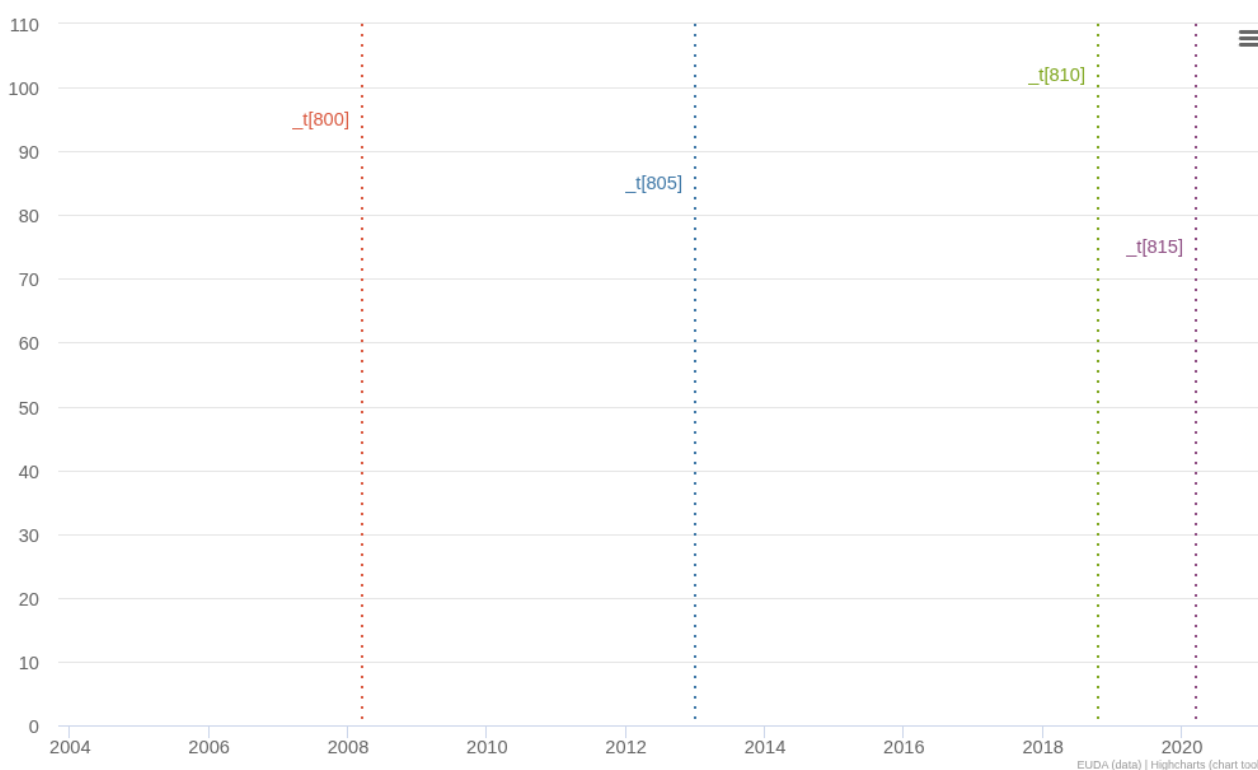
Price and purity

Although often used interchangeably, ecstasy and MDMA are not necessarily synonyms. Consumers buying ecstasy expect the tablets to contain MDMA. However, adulteration is commonplace, and therefore these tablets may contain a variety of substances in addition to, or instead of, MDMA. Rates of adulteration of ecstasy tablets, their MDMA content and their price tend to reflect broader changes in the factors that affect the MDMA market. This includes law enforcement pressure, a variety of factors affecting MDMA production, such as the availability of MDMA precursors, and, more generally, globalisation of the illicit drug supply chains.

Ecstasy adulteration is responsive to market pressures

When availability of MDMA is high, ecstasy tablets tend to contain MDMA as the main or only psychoactive ingredient, with the opposite occurring when there are shortages. These shortages tend to be transitory, which means that essentially there are cycles of adulteration. One way to observe these cycles is to monitor rates of adulteration, as expressed by the percentage of tested samples of seized ecstasy tablets that contain MDMA and no other scheduled substances. Although these data may not be representative of the market as a whole, they can provide an approximate historical overview of the changes in adulteration of ecstasy tablets in Europe (see [Figure 6.3. Indexed trend of the average percentage of ecstasy tablets containing MDMA as the only scheduled substance](#)).

Figure 6.3. Indexed trend of the average percentage of ecstasy tablets containing MDMA as the only scheduled substance



From 2004 to 2009, the proportion of seized European ecstasy tablets that contained MDMA as the only scheduled substance dropped substantially. In 2009, on average, only 31 % of the tablets analysed contained MDMA alone, compared with almost 100 % in 2004. This drop coincided with a worldwide shortage of safrole, an essential precursor for MDMA which had been under international control since 1992 but was the target of a major law enforcement clampdown around 2008. In 2009 and the following years, it was therefore unlikely that consumers who purchased ecstasy in Europe would obtain unadulterated MDMA. Crystal MDMA appeared on the European market at that time, as a response to a fall in consumer confidence due to concerns about the low quality of ecstasy. Crystal MDMA was marketed as a purer product compared to ecstasy tablets, an assertion that was and remains largely true (see Section [The trend of large tablets containing more MDMA may be reversing](#)).

As producers found designer precursor alternatives to PMK, the availability of MDMA increased. PMK glycidate, first reported in Europe in 2013, had a particular role in this resurgence (see Section [Production - designer precursors substitute scheduled MDMA starting materials](#)). The increase in ecstasy tablets containing only MDMA from 2013 until 2019 is likely to be a reflection of this, at least in part.

From 2019 onwards a new cycle appears to have begun, whereby the number of ecstasy tablets containing other substances alongside, or instead of, MDMA has increased (as expressed by a drop in percentage in the indexed trends shown above from 96 % in 2019 to 73 % in 2022). Again, these data should be interpreted with caution and not necessarily be seen as statistically representative

of the market as a whole. Nevertheless, it may be an indicator of MDMA market trends. This finding is also consistent with pressure on the MDMA market coming from reduced access to PMK glycidate (controlled internationally in 2019) and the closure of recreational settings associated with MDMA use during the COVID-19 pandemic.

Data from 10 drug-checking organisations belonging to the Trans European Drug Information (TEDI) network also support these findings: in 2019, 93 % of the 1 165 samples submitted as ecstasy contained MDMA alone, while in 2022 that percentage had dropped to 84 % (of 1 525 samples).

MDMA adulterants

Adulteration of MDMA occurs by adding other psychoactive substances, typically stimulants, to MDMA in tablet, powder or crystal form. A number of substances can also be sold as MDMA, in what is sometimes called mis-selling or adulteration by substitution – whereby another substance is passed off as MDMA to the consumer. Based on samples submitted to drug checking, one study in Spain found that mis-selling of MDMA is more frequent than adulteration: it is more common to find other substances sold as MDMA than to find other psychoactive substances added to MDMA ([Vidal Giné et al., 2022](#)). Adulteration appears to be more frequent in ecstasy tablets than in MDMA crystals.

Caffeine is the adulterant most commonly found in ecstasy tablets as well as in MDMA powders and crystals. In powders and crystals, common adulterants also include phenacetin, paracetamol or dextromethorphan ([Vidal Giné et al., 2022](#)). In tablets, adulterants typically include stimulants and MDMA-type substances – other phenethylamines with comparable subjective effects, such as MDA ⁽³⁾. In the period from 2019 to 2023, caffeine, 4-CMC ⁽⁴⁾, amphetamine, cocaine and MDA were the adulterants most commonly found in MDMA samples (tablets and powders/crystals) submitted to drug-checking organisations belonging to the TEDI network. Interestingly, synthesis by-products were also commonly reported, about which little is known and which may pose additional risks to consumers.

On occasion, neurotoxic compounds such as PMMA ⁽⁵⁾ are detected mixed with, or as a replacement for, MDMA in ecstasy tablets. Although rare, this seems to be more common in periods of low availability of MDMA and has been associated with fatal intoxications. Although MDMA alone is associated with a relatively small number of serious adverse events compared to other drugs, its health risks can drastically increase when taken in large quantities (see [Box 6.3. High-strength MDMA tablets: understanding the risks](#)) or mixed with other substances – particularly those which can synergistically increase the effects of MDMA.

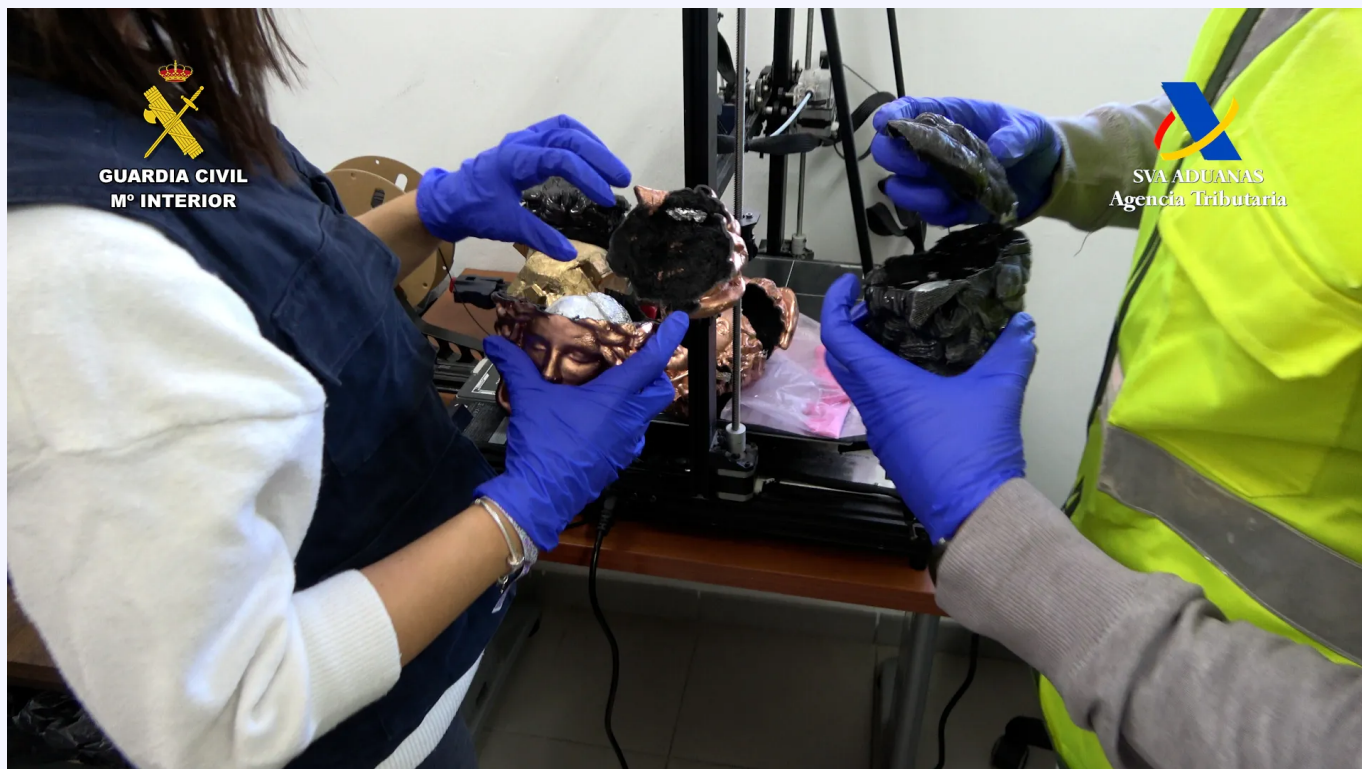
One relatively recent adulteration pattern is the increase of synthetic cathinones mis-sold as MDMA or used to adulterate MDMA. These signals were reported to the EU Early Warning System from 2022 onwards, based on identifications occurring in Spain, the Netherlands and Austria, and there are similar indications for the United Kingdom ([Pascoe et al., 2022](#)). These signals have occurred when the availability of cathinones seems to be at a historical high in the European Union (see [EU Drug Market: New psychoactive substances – In-depth analysis](#)). The detection of mixtures

of MDMA and ketamine combined with other substances including cocaine have also increased in some regions of Europe, and are sold as 'pink cocaine', 'tuci' or 'tucibi' (see [Box 6.3. What is tucibi?](#)).

Box 6.2. What is tucibi?

Tuci or tucibi (also spelled tusi and tusibi) are phonetic transliterations of '2C' (a series of psychoactive phenethylamines) or '2C-B' (one of the most well-known derivatives of that series of substances). Tuci or tucibi products can also be sold under the name 'pink cocaine' despite rarely containing 2C-type phenethylamines or cocaine. Rather, tuci and tucibi are typically pink powders, often with a sweet smell derived from food colouring (Palamar, 2023). These mixtures were first reported around 2021 in Latin America (namely Colombia) and were found to contain a variety of substances: mostly MDMA and ketamine and sometimes also cocaine, methamphetamine, opioids or new psychoactive substances. It was hypothesised at the time that drug traffickers were trying to recreate some of the properties of 2C-B by mixing substances with stimulant and hallucinogenic properties ([UNODC, 2022](#)), or to counteract some of 2C-B's less appealing features for consumers (namely it is painful to snort and damaging to the sinuses and may result in gastrointestinal distress) ([United States Department of Justice, 2001](#)).

Throughout 2021, and continuing into 2022 and 2023, a number of detections of pink powders consistent with tucibi were reported to the EU Early Warning System. The majority of these identifications were reported by Spain, although pink powders have also been reported in Czechia, Germany, Estonia, Hungary, the Netherlands, Austria, Poland, Romania, Sweden and Norway. These detections have generally been small in quantity with the powders containing a range of different substances – always ketamine and MDMA, and typically additional substances including cocaine, caffeine or cathinones. However, larger seizures of these mixtures have also been reported in the European Union. For example, in August 2022 an organised crime group was dismantled in Spain; it was involved in the trafficking of tucibi and had connections to Colombia. More than 13 kilograms of a mixture described as pink cocaine was seized, alongside 6 kilograms of cocaine, 6 kilograms of ketamine, firearms and large sums of cash and luxury items ([Guardia Civil, 2022](#)). In May 2023, a criminal network smuggling tucibi inside 3D-printed double-bottomed packages to Colombia and Australia was dismantled in Seville, resulting in nine arrests. A somewhat similar scheme was uncovered several months later, in November 2023, where 3D-printed figurines used to conceal tucibi were intercepted before being shipped to the United States ([Guardia Civil, 2022](#)).



3D manufactured figurines containing shipments of tucibi, shipped from Spain to the United States. Source: Spanish Guardia Civil.

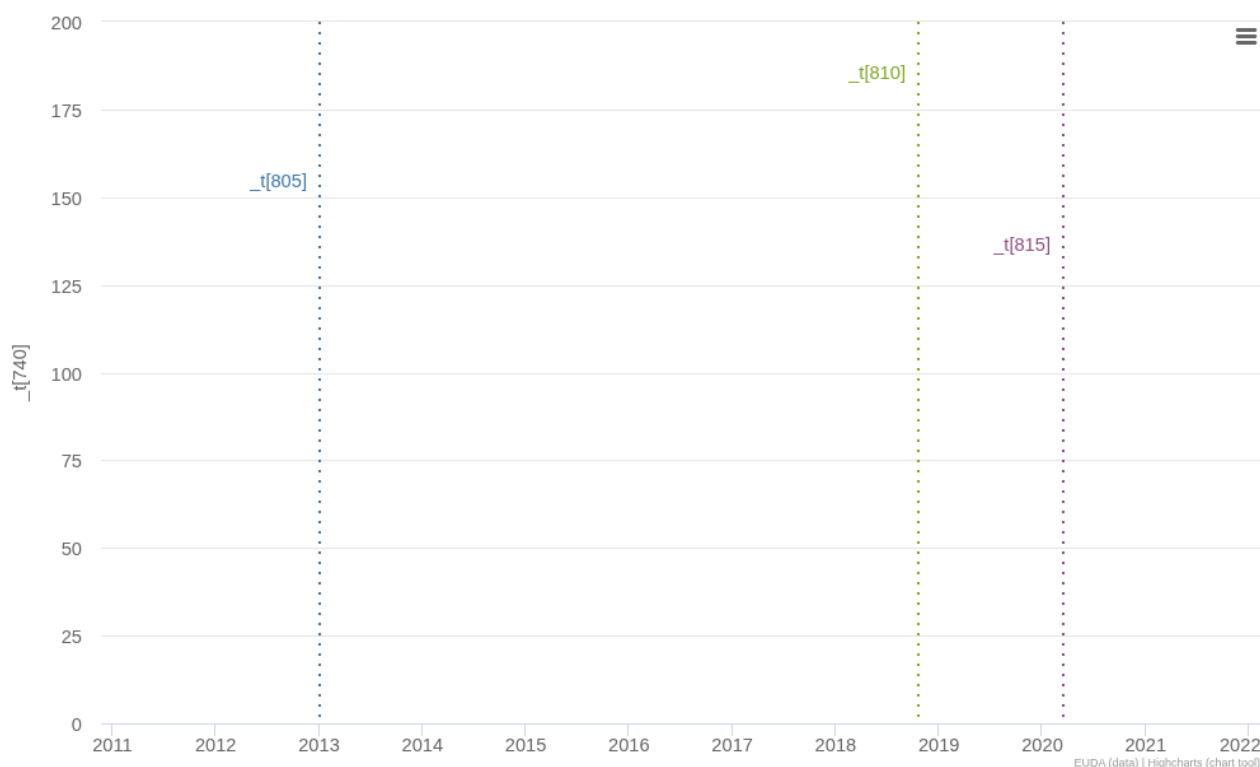
Although very rarely reported, opioids may also be found in tablets sold as MDMA. In Australia, in January 2024, potent opioids of the nitazene family were found in brightly coloured tablets with logos reminiscent of ecstasy tablets. At least three people were hospitalised after consuming these tablets.

The trend of large tablets containing more MDMA may be reversing

The average content of MDMA in ecstasy tablets appears to be on a slight downward trend from 2019, after close to a decade of increasingly stronger (and larger) tablets becoming available on the market. In 2011, ecstasy tablets contained on average between 35 milligrams and 119 milligrams of MDMA (mean: 84 milligrams). However, by 2019, those values varied between 118 milligrams and 202 milligrams (mean: 170 milligrams). This means that consumers buying ecstasy tablets in Europe in 2019 would obtain a product containing up to almost double the highest amount of MDMA recorded in ecstasy tablets in 2011.

However, from 2019 onwards, the quantity of MDMA found in tablets appears to have slightly decreased. In 2022, the values ranged between 104 milligrams and 168 milligrams (mean: 144 milligrams) – roughly comparable to values reported in 2017 (see [Figure 6.4. Quantity of MDMA, in milligrams, per ecstasy tablet \(mean\)](#)).

Figure 6.4. Quantity of MDMA, in milligrams, per ecstasy tablet (mean)



Note: Includes base and salt forms.

In the first period, between 2011 and 2019, the increase in MDMA content is largely attributed to the sale of larger (heavier) ecstasy tablets. A study conducted by 10 drug-checking organisations from the TEDI network found that the increase in MDMA content in tablets in this time period was directly related to an increase in the average weight of ecstasy tablets (Vrolijk et al., 2022). This means that the ratio of MDMA to fillers (excipients) in the mixtures used to make ecstasy tablets did not increase over this time period.

While individual tablet weight cannot be a direct predictor of how much MDMA is present in each tablet, at population level this trend is consistent when comparing larger ecstasy tablets (almost always containing more MDMA) with smaller tablets (almost always containing less MDMA). It should be noted that this trend was observed across several countries in Europe. This is consistent with law enforcement information that suggests the MDMA market in Europe is supplied by the same pool of producers, mostly based in the Netherlands and Belgium.

However, this trend appears to be changing, as ecstasy tablets have tended to contain less MDMA since 2019. This trend may be a combination of the scheduling of PMK glycidate, and a reduction in demand for MDMA during the COVID-19 pandemic, and is in line with the changes observed in the adulteration of ecstasy tablets (see Section [Ecstasy adulteration is responsive to market pressures](#)).

Data reported to drug-checking services are generally in agreement with these trends. In 2023, 39 % of the MDMA tablets tested by drug checking organisations submitting data to the TEDI

network contained between 100 and 149.0 milligrams of MDMA (35 % in the same period in 2022), and 26 % contained between 150 and 199.9 milligrams (29 % in 2022). On occasion, exceptionally high-strength ecstasy tablets were reported, reaching upwards of 250 milligrams of MDMA per tablet, which made up 1 % of all tablets tested in 2023 (3 % in the same period in 2022). High-strength MDMA tablets can have serious health consequences (see [Box 6.3. High-strength MDMA tablets: understanding the risks](#)).

Data reported to the EUDA over the last decade (2011-2021) show that while the proportion of MDMA in tablets has varied between 40 and 50 %, the purity of MDMA powder and crystals has typically varied between 70 and 80 %. Meanwhile, data from drug-checking services suggest that approximately 85 % of the powders tested in 2023 contained between 80 and 100 % MDMA (81 % in 2022).

Box 6.3. High-strength MDMA tablets: understanding the risks

Tablets containing more than 150 milligrams of MDMA are considered high-strength ecstasy tablets, given the dose of MDMA required to elicit subjective effects in adults (^a). Occasionally, tablets containing 250 milligrams or more of MDMA are encountered on EU drug markets. For example, in August 2023, tablets containing 350 milligrams of MDMA were seized in Spain. In November 2023, two tablets collected by Energy Control's drug-checking services in Spain were also of very high-strength: one contained 371 milligrams of MDMA, and the other contained 300 milligrams. In December 2023, beige-golden ecstasy pills with an Audi-logo containing over 300 milligrams of MDMA (between 316 and 324 milligrams) were submitted to drug testing in the Netherlands ([DIMS, 2023](#)). These identifications of high-strength ecstasy tablets prompted harm reduction organisations to issue alerts.

Some of the high-strength ecstasy tablets are scored to allow consumers to divide them into smaller portions. Some tablets even contain stamping that indicate the quantity of MDMA present, although this is not always accurate and may also constitute a health risk. Nonetheless, even when dividing a high-strength tablet into smaller pieces, doses may still be too high and increase the risks of acute physical and psychological discomfort (e.g. anxiety), hyperthermia and even potentially fatal reactions such as cardiac events, and serotonin syndrome (see Section [Harms and risks](#)). Harm reduction responses, including the provision of drug-checking services and appropriate risk-communication strategies, are crucial to minimise these risks.



Tablets seized in Spain containing 350 milligrams of MDMA. Source: Spanish Police.

^(a) In clinical trials, doses of MDMA capable of producing subjective effects vary from 1.5 to 1.7 mg/kg of body weight, or approximately 105 to 119 milligrams for a 70-kilogram individual. Individual factors and sex affect this dose, with women considered to be more susceptible to the 5-HT-releasing effects of MDMA (Liechti et al., 2001).

Price

Data on the retail price of MDMA in the European Union are limited, but the available indicators suggest that prices have not changed significantly over the past decade (2012-2021). For countries reporting data in 2022, a considerable retail price range was observed, with prices per ecstasy tablet ranging from EUR 4 to EUR 19 (mean) and prices of MDMA powder from EUR 6 to EUR 50 per gram. Some of the price disparity may be attributed to local factors, including distance to the producing centres (Netherlands and Belgium), but statistical artefacts (e.g. small sample sizes) may also be a factor.

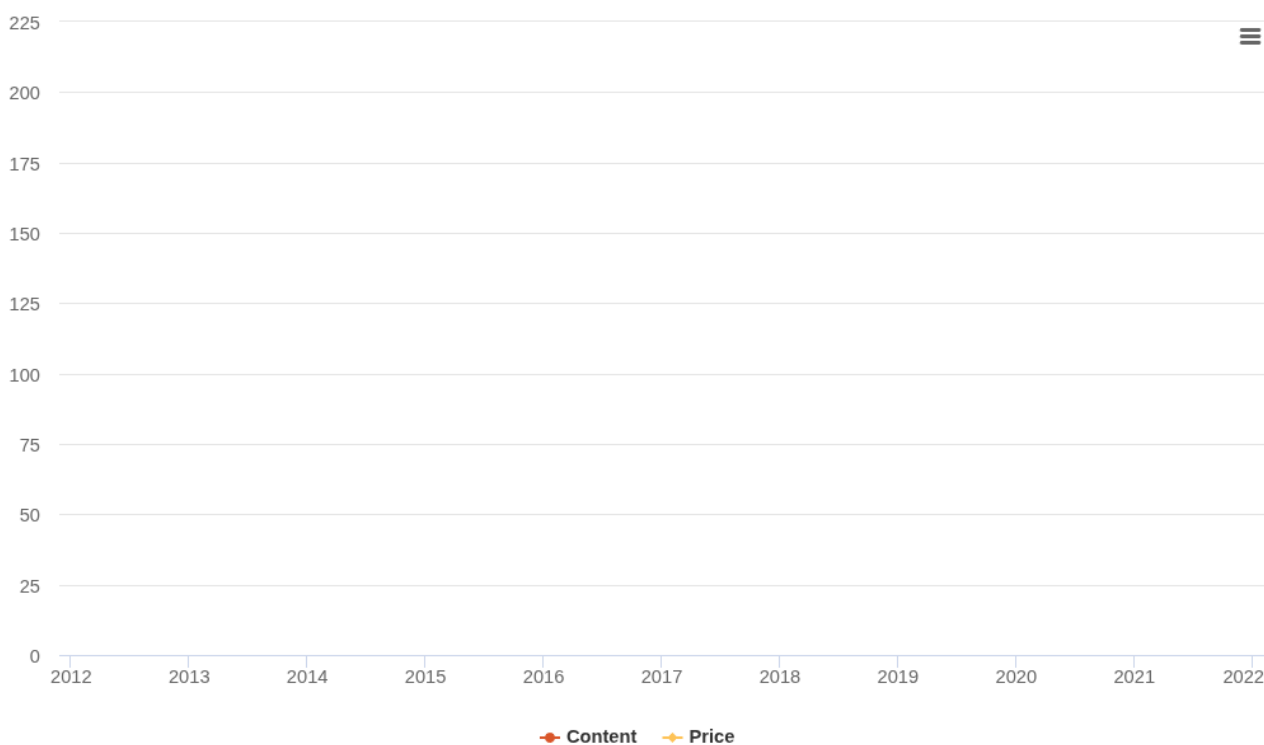
In 2021, adult respondents to the EWSD reported paying a median of EUR 7.20 per ecstasy tablet and a median of EUR 37.40 per gram for MDMA powder.

Analysis of darknet data reveals that the typical price paid is EUR 10 per gram of MDMA in tablets (a tablet containing 144 milligrams of MDMA costs on average EUR 1.44) when purchasing less than 50 tablets and EUR 10 per gram of crystals when purchasing less than 50 grams (see [Section Trafficking and distribution — online trade and distribution on darknet markets](#)).

The increase of the quantity of MDMA present in ecstasy tablets noted in the 2012-2019 period was not reflected in an increase in price per tablet (see [Figure 6.5. Indexed trends in MDMA content and price of ecstasy tablets, 2012-2022](#)). In fact, it could be argued that MDMA in the form of

ecstasy tablets became cheaper between 2012 and 2019 as tablets containing larger quantities of MDMA became increasingly available on the market (see Section [The trend of large tablets containing more MDMA may be reversing](#)). In part, this may have occurred because of lower production costs associated with the increasing availability of cheap designer precursor alternatives from China around this time. The subsequent drop in MDMA content in tablets since 2019 does not appear to have been accompanied by a significant drop in price. While a slight decrease in price was observed in 2022 for ecstasy tablets, this may not be statistically significant or sustained in the coming years.

Figure 6.5. Indexed trends in MDMA content and price of ecstasy tablets, 2012-2022



Wholesale price data on MDMA is limited. For the six countries that reported data for 2022, prices for 1 000 tablets varied from EUR 733 to EUR 7 228. In the Netherlands, in 2022, the price of 1 kilogram of (locally produced) MDMA crystals was reported at EUR 2 300, whereas 1 litre of MDMA oil (which can be converted into crystals or powder, and subsequently ecstasy tablets) was reported at EUR 1 550 per litre. Dutch law enforcement reports that during 2020, particularly during lockdown periods associated with the COVID-19 pandemic, prices of MDMA oil dropped drastically due to reduced demand. The effect appeared to have been short-lived, and by the end of 2020 the price returned to pre-pandemic values ([NPNL, 2022](#)).

Profits generated by the sale of MDMA tablets are expected to be significant given the volume of the trade. The limited literature on the subject suggests that, in 2019, the production cost of a single tablet of MDMA in the Netherlands varied between EUR 0.25 and EUR 0.40. According to Dutch Police, these tablets are sold to distributors at EUR 0.59 per unit (2022 values), who in turn

re-sell them at EUR 1.78 each (for quantities above 99 tablets). Consumers in the Netherlands will typically pay EUR 4.50 per tablet but, as stated above, prices may reach as high as EUR 19 per tablet in some European countries, and more in other regions of the world.

Prevalence and patterns of use

Historically, MDMA has been one of the most widely used illicit stimulant drugs in the European Union, associated with patterns of consumption in the context of nightlife and other recreational settings. The latest available data suggest an overall relatively stable level of consumption of MDMA in the European Union, although there are variations at national level. An estimated 12.3 million adults in the European Union (aged 15-64), or 4.3 % of this age group, have used MDMA at least once in their lifetime ([EMCDDA, 2024](#)).

Data on prevalence of use provide an indication of the structure, location and approximate size of retail MDMA markets in the European Union.

Males make up the majority of MDMA consumers in the European Union, and are three times as likely as females to report using MDMA. Males also make up the majority (68 %) of the relatively few new entrants into drug treatment in 2021 for MDMA-related problems, and the majority of patients (69 %) that experienced a hospital emergency associated with MDMA exposure in 2022 (see Section [Harms and risks](#)). This gender distribution is a commonly found pattern across other substances.

MDMA use is more likely to be reported by young adults. Surveys conducted by 26 EU countries between 2015 and 2023 suggest that 2.2 million young adults (aged 15 to 34) used MDMA in the last year (2.2 % of this age group), and 1.1 million young adults aged 15 to 24 years are estimated to have used MDMA in the last year (2.3 % of this age group) (see [Figure 6.6. Key prevalence data for MDMA/ecstasy use in Europe](#)). The age distribution of those seeking help for MDMA-related harms also reflects the higher prevalence of use among young adults: the majority of new treatment entrants for MDMA-related problems are between 15 and 24 years old and the majority of individuals presenting to hospital following an MDMA-related emergency are under 25.

According to survey data, the two largest MDMA consumer markets per capita in the European Union are the Netherlands and Ireland, with 6 % or more people aged 15 to 34 reporting to have consumed MDMA in the last year. General population surveys in many countries showed that MDMA prevalence was declining from peak levels attained in the early to mid-2000s. Of the 13 EU countries that undertook surveys since 2021 and provided confidence intervals, four reported higher estimates than their previous comparable survey, nine reported stable estimates and one reported a decrease. Where prevalence is higher, it may suggest that MDMA is no longer a niche or subcultural drug limited to dance clubs and parties in those markets. Importantly, reported use of the drug declined temporarily during periods of social distancing during the early phases of the COVID-19 pandemic, coinciding with the closure of nightlife and the cancellation of events where MDMA is typically consumed.

Wastewater data from 2023 complement the available data from general population surveys, drug treatment and hospital emergency admissions; however, wastewater is not uniformly analysed across the European Union. The data show that large quantities of MDMA are used in cities in Belgium, Czechia, the Netherlands, Spain and Portugal. However, use varies considerably across the study locations, typically being higher in larger cities compared to smaller ones. Looking at longer-term trends in wastewater analysis, in most cases the levels increased between 2011 and 2016, and have since fluctuated. In 2020, during the COVID-19 pandemic, almost half of the cities (24 of 49) reported a decrease in MDMA levels, with 18 reporting an increase. In 2021, 38 out of 58 cities reported a decrease. Of the 69 cities that have data for 2022 and 2023, 42 reported an increase, 11 reported a stable situation and 16 a decrease. All of the 10 cities with data for both 2011 and 2023 had higher MDMA levels in 2023 than in 2011 (see [Figure 6.6. Key prevalence data for MDMA/ecstasy use in Europe](#)).

Figure 6.6. Key prevalence data for MDMA/ecstasy use in Europe

This data explorer enables you to view our data on the prevalence of MDMA use by recall period and age range. You can access data by country by clicking on the map or selecting a country from the dropdown menu.

Notes

Prevalence data presented here are based on general population surveys submitted to the EMCDDA by national focal points. For the latest data and detailed methodological information please see the [Statistical Bulletin 2024: Prevalence of drug use](#).

Graphics showing the most recent data for a country are based on studies carried out between 2013 and 2023.

Prevalence estimates for the general population: age ranges are 18-64 and 18-34 for Germany, Greece, France, Italy and Hungary; 16-64 and 16-34 for Denmark, Estonia and Norway; 18-65 for Malta; 17-34 for Sweden.

Patterns of use

Polydrug use is common among MDMA users. Cannabis, alcohol and to a lesser extent cocaine, amphetamine, methamphetamine, ketamine and GHB are consumed in combination with MDMA. Cannabis is the most frequently used substance alongside MDMA, as reported by EWSD respondents in 2018 ([Gremeaux, 2022](#)) while alcohol is present in the majority of hospital emergencies associated with MDMA exposures (see Section [Harms and risks](#)).

Information gathered in the 2021 EWSD suggests that, on average, those who reported having used MDMA in the last 12 months, consumed one ecstasy tablet or 0.4 grams of MDMA in crystal or powders form each time they used the drug. While the number of tablets used per session has not changed substantially from literature findings dating from several decades ago, today's MDMA users generally ingest higher quantities of the psychoactive substance than before ([Gremeaux, 2022](#)). This is because ecstasy tablets now contain more MDMA, on average, than they did in the past (see Section [Price and purity](#)).

According to EWSD respondents, the main motivations for MDMA use are ‘to get high/for fun’ and ‘to socialise’. Wastewater data also reflect the predominant use of ecstasy in recreational settings, with more than three quarters of cities reporting that higher loads of MDMA were present in wastewater during the weekend than during weekdays in 2022. This is in accordance with literature findings, which further suggest that the substance is typically consumed in discrete episodes: over a 24-hour period (usually Friday or Saturday night) or over a whole weekend or duration of a music festival ([Cole, 2014](#)). There are rare reports of extended use of ecstasy over longer periods of time.

Market size estimate

Demand-based estimate

The basic approach used to estimate the value of the European MDMA market is to gauge the quantity of MDMA consumed in Europe and multiply it by its price (EMCDDA, 2019a). The total quantity of MDMA consumed in Europe is reached by multiplying the number of users by the amount used, combining prevalence data from general population surveys and the amount used from the EWSD.

Prevalence rates of the number of MDMA users, obtained from general population surveys, are multiplied by 2021 Eurostat population data of the 15-64 age group. Similar to the MDMA market size estimate published in 2019, it was only possible to distinguish two broad groups of MDMA users from the general population surveys, based on frequency of use, which were mapped onto the data available from the EWSD as follows:

- **infrequent users** who used MDMA less than 11 times per year, approximated in the general population surveys by those using in the last year but not in the last month (subtracting last month prevalence from last year prevalence); and
- **frequent users** who used MDMA on 11 or more days per year, approximated in the general population surveys by those using in the last month.

As such, only last year prevalence and last month prevalence data are required to distinguish these two groups. In the case of Belgium, Czechia, France, Luxembourg, Malta, Norway and Portugal, no recent data for last month prevalence of MDMA were reported, so data were imputed.

The amount of MDMA used annually by these two different user groups was estimated using data from the EWSD. The annual amounts used, in tablets, could be estimated for each survey participant based on their responses to questions on the number of tablets used on a typical day, multiplied by the number of days they said they used the drug in a year. In calculating the amounts used, it was assumed that the maximum realistic amount to use in a day was five tablets; those reporting higher values were excluded. Sharing tablets was not accounted for, which could support higher values. For some countries (Belgium, Croatia, Denmark, France, Malta, Netherlands, Norway and Türkiye) no data were available from the EWSD. In these cases, the

missing information was replaced with the mean, computed on the remaining countries.

The price data used was the average retail price for MDMA in the country concerned, drawn from the EUDA’s annual data collection (see Section [Price and purity](#)). When more than one price point was provided, the rule was to use the following: mode, median, mean, average of minimum and maximum (min-max average), in order of availability.

By combining these data, the estimated annual value of the retail market for MDMA in Europe is at least EUR 594 million, around 90 % of which can be attributed to frequent users. Estimates of amounts used suggest that about 72.4 million ecstasy tablets were consumed in the European Union in 2021.

Wastewater-based methods

Demand-based estimates may underestimate the size of drug markets for a number of reasons, including difficulty reaching some groups of users, and underreporting of prevalence and quantities used by respondents. Drug residue measurements in wastewater show promise as an alternative method to estimate city-level drug consumption ([van Nuijs et al., 2011](#)), since the amount found in waste is directly related to the quantity of drugs consumed by the population served by the wastewater treatment plant. In a study commissioned for this analysis, wastewater estimates were found to give different results, either higher or lower, than demand-based estimates, depending on the drug concerned. Estimates based on wastewater data for MDMA were noted to be more than double the amount estimated by the demand-based estimate.

When using wastewater data to estimate MDMA markets, the use of MDMA is based on the excreted parent drug, rather than on an excreted metabolite (which is the case for some other drugs, such as cocaine and heroin). This raises the possibility that this method may overestimate use if large quantities of unused MDMA are discharged directly into the sewerage system ([Bettington et al., 2018](#); [Guirguis, 2010](#)).

It was possible to construct estimates of city-level MDMA market sizes by combining wastewater data with country-level data for average retail purity and price. City-level market size estimates were developed for Brussels, in Belgium, and five cities in Sweden (see [Table 6.1. Estimated value of MDMA consumed in selected European cities, 2020](#)). While the population of Brussels is larger than that of Stockholm, the amount of MDMA consumed is almost 30 % higher in Stockholm. Furthermore, when factoring in the higher price of MDMA in Sweden, the market value in Stockholm is about 3.5 times that of Brussels. While affected by limitations and uncertainties, this type of data analysis shows great promise as a complementary information source to monitor the size and changes in drug markets ([Quireyns et al., 2023](#)).

Table 6.1. Estimated value of MDMA consumed in selected European cities, 2020

Country	City	Population	Amount of pure drug consumed (grams/year)	Amount of retail-level drug consumed (grams/year)	Value (EUR/year)
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Belgium Brussels	953 987	34 137	262 596	1 376 002
Sweden Gävle	85 000	4 377	26 856	375 980
Sweden Sandviken	28 000	826	5 069	70 967
Sweden Söderhamn	14 500	343	2 105	29 470
Sweden Stockholm	860 800	55 223	338 790	4 743 058
Sweden Uppsala	200 000	6 541	40 126	561 765

Harms and risks

MDMA consumption is associated with an increase in the activity of serotonin and dopamine in the brain, which may elicit feelings of euphoria, empathy and stimulation. Harms associated with MDMA are typically modest and transient in their impact, but serious adverse events, including those with fatal outcomes, may occur – especially when large quantities of MDMA are consumed.

MDMA use can lead to dangerous levels of hyperthermia and dehydration – often exacerbated by the environments in which it is commonly consumed, such as crowded nightclubs or music festivals. In addition, in those conditions, some people may drink too much water leading to hyponatremia or water intoxication, which can be fatal ([Parrott, 2013](#)). MDMA use may be particularly risky for individuals with pre-existing cardiac issues ([Liechti, 2003](#)). Long-term and excessive use of the substance, however rare (see Section [Patterns of use](#)), has been associated with neurotoxic effects and can result in impairments in cognitive function, depression, and anxiety ([Buchert et al., 2003](#); [Parrott, 2013](#)).

Acute MDMA toxicity is associated with serotonin syndrome (serotonin toxicity), whereby too much serotonin is present in synapses in the brain, resulting in a combination of neuromuscular, autonomic and psychologic symptoms ([Foong et al., 2018](#)), manifested in a range of mild to severe symptoms. Mild symptoms can include restlessness and tremors, whereas more serious symptoms may include severe hyperthermia and muscle stiffness, which have the potential to be life-threatening ([Makunts et al., 2022](#)). The use of MDMA in high doses (for example in ecstasy tablets with high MDMA content) and the use of MDMA alongside substances with serotonergic effects such as selective serotonin reuptake inhibitors (commonly prescribed as antidepressants), other stimulants and opioids substantially increases the risk of serotonin syndrome. Nonetheless, generally speaking, the number of drug-related deaths associated with exposure to MDMA is low compared to other illicit substances, considering the number of people that consume the drug.

MDMA use continues to represent an important issue for prevention and harm reduction messaging and interventions, not only because its toxicity can be fatal, but also because the availability of high-strength and adulterated products increases the risk of adverse health outcomes associated with its consumption ([Roxburgh et al., 2021](#)).

MDMA toxicity can result in hospital emergencies and fatalities, but incidence is low

MDMA-related deaths are relatively infrequent. However, studies have suggested that an increase has been observed in some countries. A review of MDMA-related mortality in Australia and three European countries (Portugal, Finland and Türkiye) showed an increase in the number of fatalities between 2011 and 2017 ([Roxburgh et al., 2021](#)). This increase coincided with a period of increased purity and availability of MDMA in these markets, suggesting that changes in MDMA supply may result in serious public health risks. A minority of these fatalities (13-25 %) were due to MDMA toxicity alone, with multiple drug toxicity being more prevalent.

MDMA is reported in small numbers of drug-induced deaths, and in most countries less than 1 in 20 cases involve the drug. For example, in Germany, MDMA was mentioned in 1 in 25 fatal overdoses in 2022, with most of the cases involving multiple drugs. Türkiye is an outlier, with MDMA identified in almost 1 in 5 drug-induced deaths (46 out of 246 in 2022). The majority of drug-induced deaths in Türkiye are among younger people, predominantly males, and are more likely to involve stimulant drugs rather than opioids, compared with the European Union and Norway.

In 2022, hospital emergencies associated with MDMA were registered in 19 of the 23 sentinel centres that reported data to the EUDA through the Euro-DEN Plus project. The share of acute drug-toxicity presentations involving MDMA increased as compared to other substances and the previous year (or MDMA was reported for the first time) in 10 of the Euro-DEN hospitals. In 5 hospitals, this share decreased and in 4 it remained stable. Only in the sentinel hospitals in Belgium, France and the Netherlands, was MDMA involved in more than 1 in 10 presentations.

A minority of hospitalisation cases were severe, and admission to the intensive care unit was needed in 5 % of these presentations. Half of the centres reported no MDMA-related admissions to the intensive care unit.

The median percentage of drug-related presentations involving MDMA was 4.7 % across the reporting hospitals in 2022, and 2.8 % for presentations involving MDMA as the sole drug reported. Alcohol was co-consumed across 67 % of all MDMA-related hospital emergencies, and the other drugs commonly consumed alongside MDMA included cocaine, amphetamine, methamphetamine, ketamine, heroin, GHB and cannabis.

Demographic insights obtained from drug-related emergencies are in general agreement with other data sources (see Section [Prevalence and patterns of use](#)). Emergencies mostly occurred among young adults aged below 25 years, which is the group most likely to consume MDMA. Across the 24 hospitals reporting presentations with MDMA in 2022, the median proportion of presentations aged below 25 years was 45 %. In contrast, the median proportion of presentations aged over 45 years was 4 % across the 24 hospitals. Most MDMA-related hospital emergency cases involved males, confirming trends in prevalence and patterns of use.

MDMA rarely cited as a reason for entering drug treatment in Europe

At EU and country level, people seeking drug treatment for MDMA-related problems represent a very small proportion of all clients entering drug treatment – typically less than 0.5 % of all new entrants into drug treatment in any given year. In 2022, a total of 853 clients entered drug treatment for the first time due to MDMA-related problems in 2022. Almost two thirds (503) of all new treatment entrants were reported from three countries, Germany (213); France (169) and Hungary (121). This may not reflect use patterns, but rather differences in the offer, legal requirement and access to drug treatment between countries.

The total number of clients who entered treatment for the first time due to MDMA-related problems has remained relatively stable in Europe over the years, with a slight decrease in 2020 and 2021, likely due to the disruptions to nightlife and treatment or monitoring activities at treatment centres during the COVID-19 pandemic.

(1) The [European Web Survey on Drugs](#) (EWSD) is a voluntary, anonymous survey run by the EUDA and its partners targeting directly people who use drugs who are 18 years old or older, and are living in one of the participating countries.

(2) Data reported to the Trans-European Drug Information project (TEDI) by drug-checking services in Austria, Belgium, France, Italy, Luxembourg, Portugal, Slovenia and Spain.

(3) MDA is phenethylamine also known as ‘sass’ or 3,4-Methylenedioxymphetamine. IUPAC name: 1-(2H-1,3-benzodioxol-5-yl)propan-2-amine.

(4) 4-CMC is a cathinone also known as ‘clephedrone’. IUPAC name: 1-(4-chlorophenyl)-2-(methylamino)propan-1-one.

(5) PMMA is a structural analogue of PMA (para-methoxyamphetamine) and methamphetamine. IUPAC name: *N*-methyl-1-4-(methoxyphenyl)-2-aminopropane.

References

Consult the list of [references](#) used in this module.

The data used to generate the infographics and charts on this page may be found below (CSV format). Additional information, metadata and methodological notes may be found in the [EU Drug Market: MDMA source data entry in our data catalogue](#).

Please note that prevalence and patterns of use as well as wastewater data can be found in [European Drug Report 2024: Trends and Developments: source data](#).

MDMA markets data

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- [Table EDMR2025-MDMA-15. Distribution of usual MDMA forms across countries](#)
- [Table EDMR2025-MDMA-16a. MDMA samples analysed by drug-checking services](#)
- [Table EDMR2025-MDMA-16b. MDMA samples analysed by drug-checking services by sample form, 2019-2022](#)
- [Table EDMR2025-MDMA-17. Indexed trend of the average percentage of ecstasy tablets containing MDMA as the only scheduled substance](#)
- [Table EDMR2025-MDMA-19. Indexed trends: price and MDMA content, retail \(2011=100\)](#)
- [Table EDMR2025-MDMA-20a. Prevalence of MDMA \('ecstasy'\) use in Europe](#)
- [Table EDMR2025-MDMA-20b. MDMA residues detected in wastewater in selected European cities: most recent data](#)
- [Table EDMR2025-MDMA-6.4 Quantity of MDMA, in milligrams, per ecstasy tablet \(mean\), 2004-2022](#)

Prevalence of drug use data tables including general population surveys and wastewater analysis (all substances)

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- [Table EDR24-GPS-1. Prevalence of drug use in Europe, based on most recent general population surveys \(2022 or most recent year\)](#)
- [Table EDR24-GPS-2. Prevalence of drug use in Europe, trends](#)
- [Table EDR24-WW-1 Mean weekly measurements by targeted substance from wastewater analysis in selected European cities in 2023, in](#)

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